EVIDENCE MASTER / COMPLETE PRESERVATION EDITION

FARO Cross-Architecture Physical Execution, Decision Parity and Latency Evidence

ESP32-S3 · STM32F446RE ARM Cortex-M4 · CH32V307 RISC-V
9 November 2026

Purpose

This master preserves the complete preceding 15-page physical parity report, the complete 18-page integrated technical review (including additional analyses performed during the consolidation), and the ten-section recovered-provenance addendum supplied by the owner. A final verification annex provides reproducible manifest-digest evidence and uninterrupted SHA-256 identifiers. No pages are silently dropped or rewritten.

How to read this edition

Part	Retained material	Purpose
Volume I	Original 15-page consolidated physical report	Unmodified source pages, including the 30 original ESP32-S3 R2 UART lines, historical latencies, 100K results and source registry.
Volume II	Additional 18-page integrated engineering review	Unmodified source pages retaining all added findings, provenance reconciliation, the complete 12-artifact manifest and source classification.
Volume III	Owner-supplied ten-section recovered-provenance addendum	Complete text, including distinct R2/Step20/final ESP campaigns, ARM and RISC-V chronology, bounded remaining verification steps.
Volume IV	New cross-check and digest appendix	Separately reproduced original 12-item evidence-set digest, full-hash text register and evidentiary classifications.

Evidence boundary

The three microcontroller families reported physical structural-parity PASS gates. A later owner-run comparison reported identical 49,920-byte OSH_PARITY_CASES objects in four SHA-256-identified ELF files. This static equality does not independently attest the exact loaded image of every historical session. The latency campaigns have different measurement boundaries. No new physical execution, device programming, or flash readback was performed for this master.

VOLUME I

Original Complete Physical Parity Report

15 original pages. Whole-document preservation: physical execution, 30-case matrix, 100K results, historical latency and full R2 case UART appendix.

Original source pages follow without alteration. A new section does not supersede or erase the retained record.

CONSOLIDATED ENGINEERING RECORD / PUBLIC EDITION

Cross-Architecture Physical Execution, Decision Parity and Latency Evidence

FARO / OSH — ESP32-S3 · STM32F446RE ARM Cortex-M4 · CH32V307 RISC-V

30	3	49,920 B
Common reference cases	Physical processor families	Identical compiled reference table in four archival ELFs

Executive finding

Three physical microcontroller families reported passing 30-case structural-parity gates. A subsequent owner-run, read-only ELF audit identified four original binary images whose compiled 30-case reference tables are identical byte for byte. This connects the archived test design more strongly than source-file similarity alone.

Recorded route latencies exist for all three families, but they use different firmware and timing boundaries. They are reported as historical measurements, not as a synchronized cross-platform speed ranking.

This document integrates the historical physical evidence, the original 30 printed ESP32-S3 UART records, the compiled-table audit, firmware provenance and declared limitations. No new physical experiment was run to prepare it.

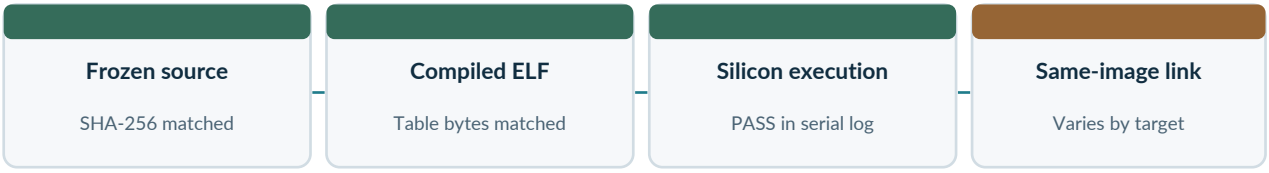
PUBLIC DISCLOSURE BOUNDARY

No private kernel source, internal memory contents, workstation paths, hardware serial numbers or private correspondence are reproduced.

01 / Executive conclusion and evidence hierarchy

What the records establish. Native FARO/OSH test code was physically executed on ESP32-S3, STM32F446RE and CH32V307, and each platform reported a successful 30-case structural-parity gate. The 30-case test is separate from each platform's later 100,000-cycle endurance and electrical experiments. [S1–S6]

What the 9 November audit adds. The operator locally identified the ARM parity, ARM 100K, RISC-V V2 and ESP32-S3 final ELF files by their full SHA-256 values. In each, the symbol `OSH_PARITY_CASES` resolved to 49,920 bytes across 30 records. All four extracted tables shared the same digest. The report reproduces the operator-supplied result; it was not independently rerun by an outside laboratory. [S7–S9]



A shared compiled table is not, by itself, a full historical Flash-image attestation.

Evidence types must not be conflated

Evidence level	What is demonstrated	Limit
Physical observation	Hardware executed the stated qualification; preserved serial PASS and other bench measurements.	Does not, by itself, identify every later archived file byte.
Compiled reference identity	Four SHA-256-identified ELF files contain byte-identical 30-case tables.	Does not, by itself, prove that each exact image was resident on the hardware during the historical capture.
Logical per-case inference	If the same verifier/corpus were executed, aggregate 30/30 success entails all five checks passed on every one of those 30 cases.	Not a substitute for 90 separately printed UART output lines.
Timing	Recorded internal route means and external oscilloscope intervals.	Measurement boundaries are not common across devices.

Evidence claim: physical structural parity with a byte-identical archived 30-case reference corpus. This is not a new three-chip rerun or an independent on-device Flash readback.

02 / Methodology: 30 cases, five causal boundaries

FARO is an embedded artificial-intelligence architecture, not merely a PID-like controller. This protocol evaluates a specified native state-to-selection/meaning pathway, not the complete cognitive system or all future deployments. [S1, S7]

Gate	Boundary	Observed condition required for PASS
L1	Endogenous final state	Numeric agreement against the frozen reference state.
L2	Settle and vortex trajectory	Path continuity, native-state evolution and omega agreement within declared tolerances.
L3	Homeostatic Attention Loop (HAL)	Expected candidate counts and selected indices; enforcement of authority boundary.
L4	Selector identity	Exact identity and ordered index of selected edges.
L5	Final semantic result	Expected final slots, fact hashes and final-meaning digest.

Acceptance logic

The examined verifier returns `run_one = L1 && L2 && L3 && L4 && L5`. The aggregate routine processes the 30 reference records and succeeds only when all 30 per-case predicates are true. Consequently, **for the exact verifier and compiled corpus that were executed**, an aggregate 30/30 PASS logically entails success on every one of the 30 cases. Printing an individual UART line is not a prerequisite for that logical implication. [S7–S9]

Numerical precision is an acceptance criterion, not a measured error distribution

Check	Threshold / condition	Interpretation
Strict state	2×10^{-5}	Declared strict state comparison.
Conditioned state	2×10^{-3}	Only for explicitly ill-conditioned trajectory cases.
Omega	2×10^{-5}	Declared trajectory angular/tangent-related comparison.
Selection / meaning	Exact identities / digests	Discrete equality; not bit-exact floating-point arithmetic.

Three different meanings of parity

PC-to-MCU: a board is checked against the frozen PC-derived reference. **Within-target structural:** all five gates pass on one platform. **Cross-target:** the board images use the same corpus and compatible checking logic and satisfy that common reference. The third claim additionally requires executed-image provenance; separate PASS summaries alone would not prove a common corpus. [S7]

03 / Compiled reference corpus: four-file identity audit

A local, owner-run audit on 9 November scanned 54 ELF candidates, selected four by exact full-file SHA-256 and extracted `OSH_PARITY_CASES` using ELF symbol-to-section addressing. Every selected table was 49,920 bytes, equating to 30 × 1,664-byte records, and their table digests were identical. [S9]

Role	ELF identity	Full-file SHA-256
ARM parity / 20K	`osh_stm32_uart.elf`	9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645
ARM 100K	`osh_stm32_100k.elf`	98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1
RISC-V V2 100K	`OSH_CH32V307_100K_V2.elf`	cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517
ESP32-S3 final	Final physical ELF	30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

COMMON COMPILED TABLE / SHA-256

`4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf`

COMMON ARCHIVED SOURCE FIXTURE / SHA-256

`23863039f2b2b920fe578c6dc6e0e00624ba34e6add00709522be8d6e70ec321`

The two digests differ by design: one identifies the header/source fixture, the other identifies the initialized data embedded in linked ELF files. The frozen ARM source fixture was also re-hashed locally on 9 November and matched the reference value. [S7–S9]

Interpretation

This closes the previously outstanding **archived compiled-corpus comparison** for the four identified ELF files. It does **not** claim that machine code, peripheral libraries or startup code are identical across ISAs, nor that a table match alone proves use by a particular historical programming session.

The original checker also reports matching whole-file ELF digests, providing an unambiguous identity of the four archived artifacts, rather than selecting same-named but different firmware revisions. [S9]

04 / ESP32-S3: R2 UART, RAM execution and final campaign

The preserved ESP32-S3 R2 serial transcript records its own firmware image SHA-256, a successful image-format check, an `esptool` RAM load of four segments, followed by execution and 30 individually printed `CASE=01` through `CASE=30` records. Every printed line reports L1–L5 PASS. This is physical MCU execution rather than a host-run emulation. The R2 boot transcript reports a CPU frequency of 160 MHz; that reading is not automatically assigned to every later ESP32 firmware revision. [S1, S2]

R2 record / output	Observed evidence
Image BIN digest	`4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811`
Execution mechanism	Physical ESP32-S3; `load_ram`; all segments transferred; device executes loaded code; UART captures parity results.
Per-case result	30 original printed case lines, each L1=L2=L3=L4=L5=PASS, with NUM, HAL, SEL, FINAL and MEANING fields.
Aggregate result	Five reference levels pass; combined 150/150 checks for Step19.
Distinct ELF in compiled audit	Final physical ELF SHA-256 `30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3`.

Important identity distinction

The **R2 RAM-loaded image** that printed the 30 case records and the separately named **final physical ELF** used in the four-ELF corpus audit are **not assumed to be the same executable**. The compiled audit proves that the final ELF has the shared table; the original R2 transcript independently shows per-case results. A direct R2-ELF-to-RAM-load identity comparison is a separate provenance link.

Later 100,000-cycle physical closed loop

The 21 September dossier documents physical reference input on GPIO3, OSH processing, PWM output on GPIO14 through an external 1 kΩ / 100 µF RC network and ADC feedback on GPIO2. It recorded 100,000 cycles, INC 5,160 / HOLD 93,358 / DEC 1,482, ADC failures 0, state failures 0, a physical gate PASS and final latch PASS. Internal mean route time was 29.525 µs (24–30 µs range). These actuator totals are **not** the 30 structural-reference decisions. [S1]

05 / STM32F446RE: parity/20K and separate 100K evidence

The archived ARM parity/20K capture reports `PARITY_CASES=30`, `L1_L2_L3_L4_L5_ALL_30_CASES=PASS` and `PC_ESP32_PARITY_GATE=PASS`. Its complete SHA-256 was recomputed locally on 9 November and matched the earlier archive. [S3, S8]

ARM artifact	Reference / interpretation
Parity/20K ELF	`osh_stm32_uart.elf` · `9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645`
Parity/20K BIN	`336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae`
Parity UART capture	`c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720` (locally reverified).
ARM 100K ELF	`osh_stm32_100k.elf` · `98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1`
ARM 100K BIN	`7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997`

What was verified on the actual target

The original ARM migration record identifies `osh_stm32_100k.elf` as the programmed firmware and documents STM32CubeProgrammer's **Download verified successfully**, reset and subsequent 100K execution. It describes 30/30 parity PASS, 100,000 closed-loop cycles, 0 ADC/state failures, 36 / 99,934 / 30 INC/HOLD/DEC and a 24.937 s run. The physical loop connects DAC PA4/A2 to ADC PA0/A0. [S3, S4]

The separate programmer log naming `osh_arm_scope.elf` is a different instrumentation image and must **not** be presented as proof that the parity/20K or 100K ELF was programmed. The operator's later 100K ELF hash also does not constitute a contemporaneous on-device Flash readback. [S4, S8]

Measurement boundary

Internal route timing: 209.500 µs minimum, 241.250 µs mean, 241.875 µs maximum, using a Cortex-M cycle counter under a declared 16 MHz timing conversion (minimum 3,352, mean 3,860, maximum 3,870 cycles). A separate oscilloscope capture recorded an approximately 226 µs instrumented interval. End-of-run reference 2,662 and ADC 2,343 imply a real 319-count residual; structural PASS is not a claim of zero tracking error. [S3]

06 / CH32V307 RISC-V: V2 physical execution and separate runs

On 8 October the CH32V307VCT6 / QingKe V4F RISC-V, reporting 96 MHz target clock, performed native 30-case structural parity followed by a 100,000-cycle qualification. The preserved 11:08–11:09 timestamped capture identifies the platform, records `STRUCTURAL_PARITY_30_30=PASS`, then 100,000 completed cycles with zero state failures and zero invalid actions. [S5]

Artifact / session	Verified or preserved evidence
V2 ELF SHA-256	`cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517`
V2 HEX SHA-256	`544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21`
V2 11:08 session	30/30 PASS; 100,000 completed cycles; actions INC 174 / HOLD 99,667 / DEC 159; state failures 0; invalid actions 0.
Earlier 00:16 session	Different timestamped UART recording; archived digest `6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d`.

Identity boundary

The historical file-selection note identifies the original V2 `FLASH_100K.hex` and the matching full digest; three archived copies share it. However, the exact same-session WCHISPTool selection/verify transcript and an on-device Flash readback for the 11:08 recovery session were not recovered. This is a **historical image-attestation gap**, not a failed parity run. [S5, S9]

Do not merge distinct RISC-V experiments

The later 9 October physical DAC-to-ADC campaign used a distinct firmware HEX (digest beginning `2db7f4db`). Its actions were 36 / 99,934 / 30, not the V2 histogram above. It supplied the 356.843 µs TIM2 mean and the 284 µs externally marked wrapper interval. These measurements belong to the 9 October workload and cannot silently be assigned to V2. [S6]

The 8 October V3 DAC-to-ADC attempt had no execution UART and was explicitly marked **NOT VALIDATED** in its checkpoint. It contributes no physical PASS assertion. No cause is inferred for that missing UART. [S5]

07 / Thirty-case reference-to-target matrix

D01–D11 identify anonymized reference decision signatures, not analog actuator commands. U = printed result in the original ESP32-S3 R2 capture; G = result implied by an aggregate 30/30 gate if the executed verifier and corpus match.

Case	Class	ESP32-S3 R2	ARM	RISC-V V2
01	D01	**PASS / U**	PASS / G	PASS / G
02	D02	**PASS / U**	PASS / G	PASS / G
03	D02	**PASS / U**	PASS / G	PASS / G
04	D03	**PASS / U**	PASS / G	PASS / G
05	D03	**PASS / U**	PASS / G	PASS / G
06	D03	**PASS / U**	PASS / G	PASS / G
07	D04	**PASS / U**	PASS / G	PASS / G
08	D04	**PASS / U**	PASS / G	PASS / G
09	D04	**PASS / U**	PASS / G	PASS / G
10	D05	**PASS / U**	PASS / G	PASS / G
11	D05	**PASS / U**	PASS / G	PASS / G
12	D05	**PASS / U**	PASS / G	PASS / G
13	D06	**PASS / U**	PASS / G	PASS / G
14	D06	**PASS / U**	PASS / G	PASS / G
15	D06	**PASS / U**	PASS / G	PASS / G
16	D07	**PASS / U**	PASS / G	PASS / G
17	D07	**PASS / U**	PASS / G	PASS / G
18	D07	**PASS / U**	PASS / G	PASS / G
19	D08	**PASS / U**	PASS / G	PASS / G
20	D08	**PASS / U**	PASS / G	PASS / G
21	D08	**PASS / U**	PASS / G	PASS / G
22	D09	**PASS / U**	PASS / G	PASS / G
23	D09	**PASS / U**	PASS / G	PASS / G
24	D09	**PASS / U**	PASS / G	PASS / G
25	D10	**PASS / U**	PASS / G	PASS / G
26	D10	**PASS / U**	PASS / G	PASS / G
27	D10	**PASS / U**	PASS / G	PASS / G
28	D11	**PASS / U**	PASS / G	PASS / G
29	D11	**PASS / U**	PASS / G	PASS / G
30	D11	**PASS / U**	PASS / G	PASS / G

Matrix interpretation. The source and ELF audits establish the common archived reference corpus. The ARM/RISC-V per-case entries are logical consequences of the reported passing aggregate gates, conditional on exact execution of the reviewed verifier and corpus; they are not individually printed UART lines. Matched per-case latency was not recorded. [S7–S9]

08 / Physical endurance and hardware I/O evidence

Test campaign	Structural parity	100K output regimes	Fault summary
ESP32-S3 final physical	30 cases × 5 levels; 150/150 PASS	INC 5,160 / HOLD 93,358 / DEC 1,482	ADC=0; state=0
STM32F446RE 100K	30/30 PASS	INC 36 / HOLD 99,934 / DEC 30	ADC=0; state=0
CH32V307 V2, 8 Oct	30/30 PASS	INC 174 / HOLD 99,667 / DEC 159	state=0; invalid actions=0
CH32V307 Diligence R1, 9 Oct	30/30 PASS (separate image)	INC 36 / HOLD 99,934 / DEC 30	ADC/state failures=0

Physical input / decision / output / feedback configurations

- **ESP32-S3:** physical B25K reference → GPIO3 ADC → native FARO path → GPIO14 PWM → external 1 kΩ / 100 µF RC → GPIO2 ADC feedback. Final physical gate and latch PASS.

- **STM32F446RE:** DAC A2/PA4 → external electrical connection → ADC A0/PA0 → native state/decision loop. 100K run finished in 24.937 s.

- **CH32V307 V2 (8 Oct):** native reference-based structural parity and deterministic 100K qualification; that checkpoint did not establish a successful physical DAC-to-ADC feedback gate. The later 9 Oct report covers a distinct DAC PA4 → ADC PA5 physical experiment.

Physical tracking errors are not structural-parity errors

The ARM 100K final reference/ADC/DAC readings were 2,662 / 2,343 / 2,336 counts; residual to ADC 319 counts. The 9 October RISC-V physical-loop report also retained a nonzero tracking residual. A passing structural gate validates the defined acceptance predicate, not necessarily ideal plant tracking. [S3, S6]

Power snapshots, only at board level

The ARM laboratory supply was observed at 4.98 V and approximately 27–38 mA (about 134–189 mW complete-board input). The later RISC-V bench showed approximately 4.82 V and 10–11 mA (about 48–53 mW complete-board indications). These are setup-specific snapshots, **not isolated OSH-core power or measured energy per decision**. [S3, S6]

09 / Historical latency, instrument boundaries and accuracy

The following are recorded physical measurements under different firmware, peripheral paths and timing boundaries. They are preserved as evidence; they are **not** a controlled three-way speed comparison. [S1, S3, S6]

Campaign / method	Min µs	Mean µs	Max µs	Measurement boundary
ESP32-S3 final 100K	24	29.525	30	Internal final closed-loop route; ADC / OSH / PWM / RC setup.
STM32F446RE 100K	209.500	241.250	241.875	Cortex-M internal cycle counter; declared 16 MHz conversion.
CH32V307 Diligence R1, 9 Oct	281	356.843	361	TIM2 native interval, PA8 marker around ADC-read-to-DAC-write wrapper.

Independent oscilloscope observations

Target	Recorded external interval	Correct interpretation
ESP32-S3	23.0 µs route; 660 ns separate action/selection interval	Two different instrumentation builds and captured windows; neither is a timing distribution.
STM32F446RE	Approximately 226 µs	Externally marked execution interval, not precisely the same boundary as the DWT timing series.
CH32V307, 9 Oct	284 µs external PA8 interval	Single oscilloscope cursor observation of a wrapper-defined ADC-to-DAC window.

No invented latency or accuracy ranking

The retrieved series do not supply P50, P95 or P99 for all three, nor timing observations for the matched 30-case corpus under a common scope. These percentiles cannot be derived from minimum, mean and maximum alone. Likewise, the numeric tolerances in section 02 are thresholds, not measured worst-case deviations, and cannot be interpreted as three-platform ADC error distributions.

A genuine matched latency study would use one ordered case/input sequence, identical initial state, a consistent operation boundary, known clocks, repeated raw cycle counters and explicit decisions on whether ADC acquisition, output writes, UART and settling are included. This document does not claim that a new study of that form occurred.

10 / Integrity anchors: firmware, fixture and captures

Archived artifact	SHA-256 (64 hexadecimal characters)
Compiled 30-case table (4 ELF audit)	`4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf`
Frozen source fixture (3 lineages)	`23863039f2b2b920fe578c6dc6e0e00624ba34e6add00709522be8d6e70ec321`
ESP32-S3 R2 RAM-loaded BIN	`4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811`
ESP32-S3 final physical ELF	`30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3`
ESP32-S3 final physical BIN	`86457e990bd1b96a093f418c5ce8f71b60d41f55c4065757df4ca543c7b4863d`
ARM parity/20K ELF	`9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645`
ARM parity/20K BIN	`336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae`
ARM parity/20K original UART	`c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720`
ARM 100K ELF	`98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1`
ARM 100K BIN	`7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997`
RISC-V V2 100K ELF	`cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517`
RISC-V V2 100K HEX	`544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21`
RISC-V V2 00:16 original UART	`6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d`
RISC-V distinct 9 Oct loop HEX	`2db7f4dba1893c67be1ebcdca9f0b0a6db43484eb9f1fb75e45301bf63bb1268`

These hashes preserve artifact identities; they are not, alone, same-session Flash-readback certificates. The 11:08 RISC-V UART recording has a different digest from the 00:16 recording. No 11:08 digest is substituted here. [S5, S9]

11 / Provenance close-out and evidentiary limits

Link assessed	Disposition	Reason
Frozen 30-case header across lineages	CLOSED FOR ARCHIVED SOURCES	Source SHA-256 identity includes ARM frozen copy reverified on 9 Nov.
30-case reference data inside four archived ELFs	CLOSED IN OWNER-RUN AUDIT	Each matches whole-file reference SHA, and each ELF table equals 49,920 bytes with the same table digest.
ESP32-S3 R2 image → original UART	SUPPORTED BY RAM-LOAD TRANSCRIPT	esptool RAM segment load/execution followed by 30 individually printed PASS cases.
ESP32-S3 R2 ↔ separately archived final ELF	NOT ASSERTED AS BYTE-IDENTICAL	One is the R2 image that printed cases; four-way ELF audit includes the distinct final physical ELF.
ARM 100K programmed image → physical runtime	DOCUMENTED HISTORICALLY	Migration record names 100K ELF and programmer verification followed by run PASS.
RISC-V V2 selected HEX → exact 11:08 image	HISTORICAL PROVENANCE; NO SAME-SESSION FLASH READBACK	Selection and matching archives plus successful UART exist, but exact WCHISPTool verification transcript/readback was not recovered.
Cross-chip, matched per-case latency	NOT TESTED UNDER COMMON BOUNDARY	Three valid but different benchmark methods and windows.

Defensible publication statement

The FARO/OSH native decision path was physically exercised on ESP32-S3, STM32F446RE ARM Cortex-M4 and CH32V307 RISC-V microcontrollers. Each platform's preserved campaign reported successful 30-case structural parity. A later local binary audit identified four historical ELFs containing byte-identical 30-case compiled reference tables. The evidence supports cross-architecture structural decision-reference parity within the tested path, subject to the recorded provenance qualifications. Historical route latencies are available but are not comparable as a single matched ranking.

Beyond the evidence presented

This does not establish a new jointly rerun 90-UART-line experiment, complete same-session Flash-image attestation for every target, a matched per-case speed/accuracy benchmark, independent third-party replication, certified safety, general intelligence, consciousness or portability of every FARO subsystem. These limits do not negate the physical runs that were carried out. [S1–S10]

A / Original ESP32-S3 R2 per-case UART records

The following 30 records are transcribed verbatim from the preserved owner-supplied ESP32-S3 R2 serial transcript. They contain the individual five-level results, number-policy indicator, candidate counts and anonymized meaning hashes. They are **not** ARM or RISC-V UART lines. [S2]

```
CASE=01 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=3 SEL=3 FINAL=2 MEANING=c9bacc8972629cd6
CASE=02 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=03 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=04 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=05 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=06 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=07 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=08 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=09 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=10 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=11 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=12 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=13 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=14 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=15 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=16 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=17 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=18 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=19 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=20 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=21 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=22 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=23 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=24 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=25 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=26 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=27 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=28 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=29 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=30 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
```

Archive interpretation. All cases 01–30 report L1=L2=L3=L4=L5=PASS. The printed 64-bit MEANING field refers to a structural outcome digest, not a language response or an analog actuator action. The original capture also reports its aggregate parity gate PASS.

B / Direct physical gate excerpts and binary comparison

ARM parity/20K capture — original acceptance lines

```
OSH_ARM_EXECUTION_BEGIN
BOARD=NUCLEO-F446RE
CPU=CORTEX-M4
PARITY_CASES=30
PARITY_EXECUTION_BEGIN
PARITY_EXECUTION_END
L1_L2_L3_L4_L5_ALL_30_CASES=PASS
PC_ESP32_PARITY_GATE=PASS
OVERALL_PAPER_CLOSEOUT=PASS
OSH_ARM_EXECUTION_END
```

RISC-V V2 recovery UART — 8 October, 11:08–11:09

```
BOARD=CH32V307VCT6  |  ARCH=RISC-V  |  CPU=QINGKE_V4F
PARITY_CASES=30
PARITY_EXECUTION_BEGIN
PARITY_EXECUTION_END
STRUCTURAL_PARITY_30_30=PASS
GATE_5_OSH_STRUCTURAL_PARITY=PASS
ENDURANCE_CYCLES=100000
RISCV_100K_PROGRESS=100000/100000
STATE_FAILURES=0  |  INVALID_ACTIONS=0
ACTION_INC=174  |  ACTION_HOLD=99667  |  ACTION_DEC=159
RISCV_100K_QUALIFICATION_GATE=PASS
OVERALL_RISCV_V2_100K=PASS
```

9 November local comparison — reported results

Four exact ELF SHA-256 matches; table parser reports 30 cases, 49,920 bytes and identical 64-character table SHA-256 for ARM parity, ARM 100K, RISC-V V2 and ESP32-S3 final. The owner supplied the terminal output; this report did not independently rerun the checker or read firmware from the physical devices.

C / Source register and reproducibility notes

The following documents and records are the basis of the consolidated claims. Titles are descriptive public aliases where archive filenames include private third-party references. Original raw UART and proprietary binaries remain in the owner's custody.

Ref	Source / recorded scope
[S1]	OSH Physical Validation of a Homeostatic Cognitive Architecture, 21 September 2026. ESP32-S3 Step19/Step20, closed-loop test, external oscilloscope, image hashes.
[S2]	Original ESP32-S3 Step19/R2 physical UART transcript, September 2026. RAM-load sequence and 30 individual CASE entries; preserved in owner-provided test logs.
[S3]	ARM Physical Validation, NUCLEO-F446RE, 3 October 2026. Parity, 100K, latencies, oscilloscope, power and artifact hash ledger.
[S4]	ARM Physical Migration Completed, 3 October 2026. STM32CubeProgrammer 100K write/verify/reset and subsequent physical execution.
[S5]	RISC-V Technical Physical Validation Checkpoint, 8 October 2026; plus the later 11:08–11:09 UART recovery record.
[S6]	RISC-V Physical Benchmark and Control Evidence, 9 October 2026. Separate DAC-to-ADC native test, scope interval and board-power observations.
[S7]	FARO 30-case Source Parity Reconstruction, 8 November 2026. Original fixture identity, five-level source predicates, anonymous decision classes.
[S8]	Operator SHA-256 checks, 9 November 2026. ARM frozen fixture and ARM parity UART archive integrity.
[S9]	Owner-run read-only ELF corpus comparison, 9 November 2026. Four whole-file SHA-256 matches; 54 ELF candidates inspected; OSH_PARITY_CASES symbol extraction and four-way 49,920-byte match.
[S10]	Prior consolidated FARO/OSH Physical Parity technical evidence dossier, 9 November 2026; plus the user-supplied Cross-Architecture Physical Execution, Decision Parity and Latency Evidence narrative.

Reproducibility and confidentiality

The public model is evidence-oriented: reviewers can inspect hash anchors, corpus-extraction results, declared acceptance logic, physical-board identity, serial gates and instrumentation records without publication of the proprietary FARO kernel or stored memories. Independent replication requires separately agreed access and independently obtained measurements.

Document generated from owner-supplied historical records and the owner's new technical narrative. It is a consolidation, not a new physical measurement or certification.

VOLUME II

Integrated Engineering Verification Review

18 original pages. Preserves the additional consolidation, provenance status matrix, 12-item manifest, full SHA-256 register and further engineering analysis.

Original source pages follow without alteration. A new section does not supersede or erase the retained record.

CONSOLIDATED PHYSICAL EVIDENCE / INTEGRATED PROVENANCE REVISION

FARO Cross-Architecture Physical Execution, Decision Parity and Latency Evidence

ESP32-S3 · STM32F446RE ARM Cortex-M4 · CH32V307 RISC-V

30	3	49,920 B
Common reference cases	Physical MCU families	Compiled table in each of four ELFs

Executive finding

Archived records document native FARO structural-parity gates passing on three physical microcontroller families. A subsequent owner-run audit reports that four SHA-256-identified ELF images contain an identical **30 × 1,664-byte** compiled reference table. The 9 November review also confirms the frozen ARM fixture hash and the original ARM parity UART digest. [E4, E8, E9, E10]

This revision integrates recovered ESP32-S3 final-campaign provenance, the distinct R2 failure history, ARM and RISC-V recording chronology, physical endurance measurements, the exact 30-line R2 UART appendix and a **new independent recomputation of the final ESP32-S3 12-item evidence-manifest digest**.

No new physical test, firmware build, flash operation or device-memory readback was carried out for this edition. Recorded physical execution and later documentary verification remain distinct evidence types.

PUBLIC DISCLOSURE BOUNDARY
This report contains evidence identifiers, non-confidential measurements, acceptance semantics and anonymized per-case UART fields. It does not reproduce proprietary source code, private memory, private workstation paths, device serial numbers or correspondence with third parties.

Consolidated engineering report · 9 November 2026 · Prepared for engineering review and publication at www.oshsystems.com

01 / Executive conclusion and evidence hierarchy

What is established. Physical firmware executed the declared FARO/OSH state-to-selection path on ESP32-S3, STM32F446RE and CH32V307, and each target reported a passing structural-parity gate for 30 reference cases. The later long-running experiments are separate qualifications, not 90 independent, synchronized UART printouts. [E4–E7]

What the compiled-image audit adds. Four archived ELF images, selected by full-file SHA-256 after a local review of 54 candidate ELF files, yielded the same **49,920-byte** `OSH_PARITY_CASES` table. The table was resolved by symbol-to-section mapping, not by assuming a virtual address was an on-disk file offset. This is stronger than source-level filename similarity. [E8]

What the new provenance review adds. The final ESP32-S3 campaign is now described as containing its own 30-case gate and successful 100K run. R2 has its own verified RAM-load transcript and a separate **failed** 100K gate, despite passing 30-case structural parity. The ARM record identifies a 100K image in the write/verify/reset sequence; the RISC-V 00:16 and 11:08 runs are distinct. [E2, E3, E5–E7, E14]

Evidence layer	Evidence in the archive	Interpretation
Common reference	Identical source fixture digest across lineages; identical ELF table bytes in four identified images.	Shared frozen test data in the archived sources/binaries.
Physical execution	ESP32-S3, ARM and RISC-V runtime or UART gates report PASS on the declared protocol.	Historical chip-level execution; not a simulated three-way rerun.
Verifier semantics	Each test case passes only if L1, L2, L3, L4 and L5 pass.	Conditional case-by-case logical implication from 30/30.
Image-to-session link	RAM-load records, source manifest, programmer-console sequences and session logs.	Strength varies by target; a missing full hash-bound RAM-loader/Flash-readback link is identified.
Timing	Recorded internal-route ranges and external instrument snapshots.	Different method/firmware boundaries; no common-clock speed ranking.

Publication formulation. The available record supports **physical structural decision-reference parity for the tested native path**, with identical compiled reference tables across four identified archival ELFs. It does not establish full same-session executable attestation on every target or a matched three-chip per-case latency experiment.

02 / Tested scope and five-level acceptance logic

FARO is an embedded artificial-intelligence architecture, not merely a PID controller. The present evidence tests a specified native state-to-decision and meaning path, along with separate physical sensor/actuator qualifications. It is not a test of all FARO cognitive capabilities, AGI, certified safety or universal portability.

Gate	Causal boundary	Acceptance predicate
L1	Endogenous final state	Final state agrees with frozen reference within the declared numeric tolerance.
L2	Settle / vortex trajectory	Continuity, native settled state, trajectory and omega agree under permitted tolerances.

Gate	Causal boundary	Acceptance predicate
L3	Homeostatic Attention Loop	Expected candidate count, selected indices and authority boundary remain correct.
L4	Selector identity	Exact selected edge identity and index agree with the reference.
L5	Final semantic result	Expected final slots, fact hashes and final meaning digest agree.

What an aggregate 30/30 PASS implies

The inspected predicate is `run_one = L1 && L2 && L3 && L4 && L5`; the outer routine iterates over 30 cases and returns success only when all 30 pass. For **that exact executed verifier and matching corpus**, a 30/30 result entails five passing checks for each of the 30 cases. The individual checks need not each be separately printed on UART for the implication to hold. Binding the verifier and corpus to the tested image remains necessary. [E8, E13]

The reviewed ARM per-case source and a preserved RISC-V version differ by 462 characters of additional case logging; the other inspected per-case text matches. RISC-V R1 compiled disassembly separately supports a 30-case iteration and an aggregate count-30 success test. These are useful source/compile lineage checks, but do not substitute for the exact V2 machine-code execution attestation. [E13]

Numeric tolerance is not measured tracking accuracy

The source predicate specifies strict state tolerance **2×10^{-5}**, conditioned state tolerance **2×10^{-3}** only for explicitly ill-conditioned trajectory cases, and omega tolerance **2×10^{-5}**. Selection and semantic digests use exact discrete comparisons. These are acceptance thresholds—not observed maximum error distributions, bit-exact floating-point results or ADC tracking tolerances. [E13]

PC-to-MCU parity, within-platform structural parity and cross-target parity are different claims. The third requires a shared reference corpus, compatible predicates and credible executed-image provenance. Identical aggregate PASS strings without that context would not suffice.

03 / Four archival ELFs: compiled corpus identity

On 9 November the owner-run checker selected four exact SHA-256-matched executables and extracted initialized `0SH_PARITY_CASES` bytes using ELF symbol and section metadata. Each resolved to **30 records × 1,664 bytes = 49,920 bytes**. The reported extracts were identical byte for byte. This report reviews the retained terminal output and checker source; it did not receive or independently rerun the proprietary ELF images. [E8]

ARM parity / 20K ELF	`9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645`
ARM 100K ELF	`98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1`
CH32V307 RISC-V 100K V2 ELF	`cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517`
ESP32-S3 final physical ELF	`30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3`

Two distinct cryptographic anchors

Common compiled table · 49,920 bytes	`4a77195717d3f87585af84f5e6c0d2d8 312ed3cbcb1aaf2f7984ab7405157baf`
Common frozen source fixture	`23863039f2b2b920fe578c6dc6e0e006 24ba34e6add00709522be8d6e70ec321`

The hashes are different by design: one covers a source-header file; the other covers the compiled initialized table. The frozen ARM header was rehashed on Windows on 9 November and matched the common source digest. Whole-image ELF hashes do not need to match across different instruction sets.

Closed in the owner-run static audit: identity of the 30-case compiled data across these four archival ELFs.
Not closed merely by this comparison: whether each exact ELF/derived program image was the one executed in every historical session.

04 / Newly verified: ESP32-S3 final evidence-set digest

The original 21 September ESP32-S3 final GPIO3 manifest contains 12 ordered artifact records, including module files, build configuration, the final BIN and the final ELF. Its source-generation script has also been recovered. [E1]

For this revision the author of the consolidation independently repeated the manifest-generation algorithm on the recovered JSON **without needing access to proprietary firmware bytes**. Each record contributes `label|sha256|byte_length` in the preserved order. Records are joined with the literal two-character separator `\n` (backslash followed by the letter n), exactly as implemented in the original script, before SHA-256 is applied to the encoded string.

Recomputed evidence-set SHA-256 Matches the historical manifest digest exactly.	`d38648587bc627409e770f5b3fd3b297 2e9abe31094db8550257fc8c9a6a0106`

Manifest item	Recorded size	Identity / use
Final GPIO3 firmware BIN	202,784 bytes	SHA-256 86457e99…7c4863d. Distinct from R2.
Final GPIO3 firmware ELF	3,432,132 bytes	SHA-256 30cbbf62…f1ff3. Included in the four-ELF table audit.
Final main source	29,910 bytes	SHA-256 331c7c52…7c5000.
Native core / HAL / Vortex / Selector / compact graph	Five separately hashed files	SHA-256 values preserved in the original 12-item ledger; code not disclosed.
Build metadata	CMake, sdkconfig and defaults	Four project/configuration records in the 12-item ordered manifest.

This **closes reproducibility of the manifest digest from its recorded identifiers**. It does **not** independently rehash each original BIN/ELF, authenticate the date of physical loading, or attest device memory. The known final RAM-loader record is still not recovered with a hash-bound association to the final BIN. [E1, E14]

05 / ESP32-S3 R2: original RAM load, parity PASS, 100K FAIL

An original R2_MEANING_FIX console record (submitted 21 September at approximately 02:00 UTC) documents an `esptool.py v4.12.0 load_ram` operation: four segments were transferred and execution began at the recorded entry point. The physical ESP32-S3 then printed **CASE=01 through CASE=30**; each line reported L1–L5 PASS. That establishes the structural parity of the R2 runtime. This is a RAM execution record, not a persistent application Flash programming claim. [E2]

R2_MEANING_FIX BIN · recorded SHA-256
202,784-byte BIN · recorded full original identifier.

`4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811`

R2 result	Recorded value
Structural parity	30 cases; five levels each; all PASS.
Physical 100,000-cycle gate	FAIL (the historical failure must not be recast as a pass).
Action histogram	INC 0 / HOLD 100,000 / DEC 0.
Internal-route timing	Min 25 µs / mean 29.329 µs / max 39 µs.
Physical final latch	FAIL; failure GPIO asserted.
Execution boundary	RAM_ONLY, MOTHER_FLASH_WRITE=FALSE; UART capture follows the recorded load.

The original R2 text also includes a preceding Step20 startup and reset/watchdog events. These must be segmented by load and boot boundaries; treating the entire paste as one uninterrupted run would be misleading. The 30-line R2 record is reproduced verbatim in Appendix A. [E2]

Why retaining the failed run matters

Structural parity and an action/occupancy endurance gate test different properties. R2 preserved correct reference-case output but did not exercise all three physical action regimes during the 100K run. The later final GPIO3 campaign passed both structural and physical gates. [E2, E3]

06 / ESP32-S3 final GPIO3: 30-case gate and 100K PASS

The newly recovered final runtime account, submitted 21 September 2026 at approximately 21:02 UTC, reports a **30-case, five-level structural-parity sequence followed by a successful 100,000-cycle physical run in the same final campaign**. The owner-supplied provenance addendum describes individual case outputs for that final session. This revision corroborates its final ELF/BIN identities against the original primary manifest and the published physical paper; the full raw final RAM-loader transcript itself has not been recovered for direct reinspection here. [E1, E3, E9, E14]

Final campaign field	Recorded observation
Processor	ESP32-S3, Xtensa LX7; reported CPU clock 160 MHz; ESP-IDF v5.4.4.
Build string	21 Sep 2026, 17:20:03 (timezone not authenticated).

Final campaign field	Recorded observation
Execution boundary	EXECUTION=RAM_ONLY; MOTHER_FLASH_WRITE=FALSE.
Structural gate	Five levels, 30 cases; passing final structural gate.
Physical run	100,000 cycles, PHYSICAL_100K_GATE=PASS; FINAL_LATCH=PASS.
Action histogram	INC 5,160 / HOLD 93,358 / DEC 1,482.
Faults	ADC failures 0 / state failures 0.
Internal route	Min 24 µs / mean 29.525 µs / max 30 µs.
Final BIN / ELF	202,784-byte BIN / 3,432,132-byte ELF, identities in the original manifest.

Physical loop and retained software boundary

Physical B25K reference → GPIO3 ADC → native OSH path → GPIO14 PWM → external 1 kΩ / 100 µF RC network → GPIO2 feedback ADC. The manifest records the final firmware BIN and ELF; the 9 November static audit confirms the same compiled corpus in the final ELF. The runtime ELF-hash display contains a zero placeholder, so it cannot authenticate the loaded image bytes. [E1, E3, E9]

Precise remaining gap: a raw, final-GPIO3 `load_ram` transcript binding the bytes actually transferred to the final BIN SHA-256 was not recovered. The R2 loader record cannot be substituted for the final build. This does not invalidate the documented final run.

07 / ESP32-S3 artifact groups: no false hash conflict

Artifact group	Original identity	Meaning / distinction
Step20 stability FINAL	ELF SHA 41bbe703…28c58540; BIN SHA 4b79d07b…a23b7a0d8; BIN 287,792 bytes.	Independent Step20 stability build; not final GPIO3 firmware.
R2 meaning fix	BIN SHA 4440dc05…9aed3811; BIN 202,784 bytes.	RAM-loaded original transcript; 30-case PASS; 100K FAIL.
Final GPIO3 closed loop	ELF SHA 30cbbf62…f1ff3; BIN SHA 86457e99…7c4863d; BIN 202,784 bytes.	Final 30-case + 100K PASS campaign; manifest reproduced.

Matching BIN sizes do not imply matching BIN content. Differences among these hashes identify **different experimental artifacts**, not contradictions for a single image. Similarly, repeated labels such as R2 must be disambiguated by the project name, digest and session boundary. [E1–E3, E14]

What the final primary manifest does and does not establish

The original TXT and JSON manifests list 12 artifacts with complete recorded SHA-256 fingerprints. The JSON and source script yield a reproducible set digest; that is positive archive consistency evidence. Yet a build configuration, a hash entry and a runtime output do not themselves prove that an independently authenticated identical image occupied the processor at a historical instant. A present-day Flash readback cannot reconstruct the contents of an earlier RAM-loaded session.

08 / ARM STM32F446RE: parity/20K and distinct 100K

The historical NUCLEO-F446RE parity/20K UART archive records PARITY_CASES=30, L1_L2_L3_L4_L5_ALL_30_CASES=PASS and PC_ESP32_PARITY_GATE=PASS. The operator recomputed its SHA-256 on 9 November; it matched the October reference. Static review of the archival parity ELF/BIN found matching load bytes, a call to the per-case verifier and a 30-case all-pass path. These findings apply to that parity/20K image. [E4, E10]

ARM parity/20K UART (reverified)	c58ec37b2995e00e1f71f66659d5c112 2d613fe1d959780c9524d5e5a4b63720
ARM parity/20K BIN	336788116049fcfcbee03210c48b32ca b92c5670216e59765280c016440fcaae

A separate primary 3 October programmer/console sequence, described in the recovered addendum and original migration roadmap, identifies osh_stm32_100k.elf, STM32CubeProgrammer v2.23.0, device STM32F446xx, Flash address 0x08000000, a successful programming verification, software reset and subsequent parity/100K runtime output. This is stronger historical programming-to-runtime documentation than an unnamed programming success. [E5, E10, E14]

ARM 100K ELF (subsequent archival digest)	98a420f518d51bbd2d7a797d87df8b2c f825189181d1d7c53ada6ec1d2b95aa1
ARM 100K BIN (subsequent archival digest)	7ae036ef3983fdbf72081904de591632 fd830fc6834ee7aae47ab7d2eb212997
ARM 100K UART (later frozen register)	7d40efbca2b4393c697f128bfce78b32 881692188b11aa0b798685e57e57ef11

The ARM 100K runtime reports 30/30 parity, 100,000 cycles, 0 ADC/state failures, actions 36 / 99,934 / 30, elapsed 24.937 s and overall PASS. Its DAC A2/PA4 → external electrical path → ADC A0/PA0 loop is a physical qualification. The later 6 October full-file hashes identify archived files, but are not contemporaneous hashes embedded in the 3 October programming log. [E5, E10]

A separate programmer trace names osh_arm_scope.elf. It documents a different instrumented image and cannot authenticate the parity/20K or 100K firmware. No verified on-device readback of the 3 October 100K image is asserted.

09 / RISC-V CH32V307 V2: original and recovery runs

The 8 October V2 archive identifies native execution on CH32V307VCT6 / QingKe V4F (96 MHz), structural parity PASS over 30 cases and a 100,000-cycle qualification. A primary timestamped UART capture is reported from **00:16:16.097 to 00:16:50.425 (UTC−03)**, including CAPTURE_COMPLETE=YES. A separate recovery UART was recorded at approximately 11:08–11:09 the same day. Both report 30/30 PASS and the 100K V2 result. They are not the same byte sequence. [E6, E7, E11, E14]

RISC-V V2 100K ELF	cb9f88f38e755cb82f51fdfe0bf8000f f0963eb774195e1b3b965ae1975c1517
RISC-V V2 100K HEX	544338ec6e5a4fa4251d8dea7f557074 418c8014ea297fd2cebaa02f9134ad21
RISC-V V2 original 00:16 UART digest	6e01a89c2d17af83c2527e549e142038 ae0e66816721b8eaea2b45cc454b7b1d

RISC-V V2 field	Recorded output
Structural gate	STRUCTURAL_PARITY_30_30=PASS; GATE_5_OSH_STRUCTURAL_PARITY=PASS.

RISC-V V2 field	Recorded output
Cycles	100,000; last progression checkpoint 100000/100000.
State / action errors	STATE_FAILURES=0; INVALID_ACTIONS=0.
Action histogram	INC 174 / HOLD 99,667 / DEC 159.
Overall	RISCV_100K_QUALIFICATION_GATE=PASS; OVERALL_RISCV_V2_100K=PASS.
Two session times	Original 00:16; later recovery 11:08 (both 8 October, UTC−03).

Programming evidence: known and unknown

Recovered hash output identifies the V2 ELF and HEX. The original owner selected the V2 HEX; three archived copies match its full digest. A contemporaneous description of an erase/program/verify screenshot exists, but the original screenshot was not independently re-inspected for this revision. A definitive same-session WCHISPTool image-selection/verification log and a hash-bound on-device Flash readback have not been recovered. [E6, E11, E14]

A reused UART filename does not transfer the 00:16 digest to the later 11:08 file. The name `FLASH_100K.hex` was also reused later for a different image, so the V2 identity must remain the full digest 544338ec…f9134ad21, not the filename alone.

10 / Separate RISC-V campaigns and the excluded V3 attempt

The 9 October Diligence R1 campaign used a different native physical DAC-to-ADC firmware and protocol. Its 100K action histogram was **36 / 99,934 / 30**, whereas 8 October V2 recorded **174 / 99,667 / 159**. Those counts are not a discrepancy for one firmware; they refer to different images and experiments. [E11, E12]

RISC-V distinct 9 October physical-loop HEX	`2db7f4dba1893c67be1ebcdca9f0b0a6db43484eb9f1fb75e45301bf63bb1268`

The 9 October physical-loop probe recorded DAC/ADC level pairs 512/520, 2,048/2,048 and 3,584/3,571 (12-bit codes), meeting its declared monotonicity and maximum-128-count deviation gate. The bench used PA4/DAC1 → external connection → PA5/ADC_IN5, while host FARO V2 versus fixed PID experiments over that physical plant are a **separate host-controller boundary**; they must not be described as those host Python controllers running natively on the MCU. The Diligence R1 route timing is 281/356.843/361 µs and the external PA8 interval is 284 µs. [E12]

V3 validation status

An attempted RISC-V V3 DAC-to-ADC branch was compiled/programmed, but the 8 October checkpoint did not record an executable UART result. It remains **NOT VALIDATED** and supplies no physical PASS to the present parity comparison. The archive does not prove the cause of the missing UART; a damaged cable is not asserted as an independently established finding. [E11]

11 / 30-case reference-to-target evidence matrix

D01–D11 are anonymized reference classes (selector/final-meaning signatures), not INC/HOLD/DEC commands. The R2 ESP32-S3 column is directly printed UART (**U**). A later final ESP32-S3 30-case gate is reported in the recovered provenance account (**F**). ARM and RISC-V case-wise PASS values follow logically from their aggregate gate when the executed verifier/corpus linkage holds (**G***). [E2, E3, E4, E6–E8, E13–E14]

Case	Class	ESP R2	ESP final	ARM	RISC-V V2
01	D01	PASS / U	PASS / F	PASS / G*	PASS / G*
02	D02	PASS / U	PASS / F	PASS / G*	PASS / G*
03	D02	PASS / U	PASS / F	PASS / G*	PASS / G*
04	D03	PASS / U	PASS / F	PASS / G*	PASS / G*
05	D03	PASS / U	PASS / F	PASS / G*	PASS / G*
06	D03	PASS / U	PASS / F	PASS / G*	PASS / G*
07	D04	PASS / U	PASS / F	PASS / G*	PASS / G*
08	D04	PASS / U	PASS / F	PASS / G*	PASS / G*
09	D04	PASS / U	PASS / F	PASS / G*	PASS / G*
10	D05	PASS / U	PASS / F	PASS / G*	PASS / G*
11	D05	PASS / U	PASS / F	PASS / G*	PASS / G*
12	D05	PASS / U	PASS / F	PASS / G*	PASS / G*
13	D06	PASS / U	PASS / F	PASS / G*	PASS / G*
14	D06	PASS / U	PASS / F	PASS / G*	PASS / G*
15	D06	PASS / U	PASS / F	PASS / G*	PASS / G*
16	D07	PASS / U	PASS / F	PASS / G*	PASS / G*
17	D07	PASS / U	PASS / F	PASS / G*	PASS / G*
18	D07	PASS / U	PASS / F	PASS / G*	PASS / G*
19	D08	PASS / U	PASS / F	PASS / G*	PASS / G*
20	D08	PASS / U	PASS / F	PASS / G*	PASS / G*
21	D08	PASS / U	PASS / F	PASS / G*	PASS / G*
22	D09	PASS / U	PASS / F	PASS / G*	PASS / G*
23	D09	PASS / U	PASS / F	PASS / G*	PASS / G*
24	D09	PASS / U	PASS / F	PASS / G*	PASS / G*
25	D10	PASS / U	PASS / F	PASS / G*	PASS / G*
26	D10	PASS / U	PASS / F	PASS / G*	PASS / G*
27	D10	PASS / U	PASS / F	PASS / G*	PASS / G*
28	D11	PASS / U	PASS / F	PASS / G*	PASS / G*
29	D11	PASS / U	PASS / F	PASS / G*	PASS / G*
30	D11	PASS / U	PASS / F	PASS / G*	PASS / G*

G* = gate-implied conditional on exact verifier and corpus executed. U = original printed R2 record. F = final-session passing gate described in the recovered evidence addendum; this edition does not substitute the R2 raw lines for final raw

lines. This is not 90 independently captured line-by-line triplets or 30 matched per-case latency measurements.

12 / Physical runs, reference inputs and actuator results

Campaign	100K actions INC / HOLD / DEC	Error / gate summary
ESP32-S3 final GPIO3	5,160 / 93,358 / 1,482	ADC_FAIL 0; STATE_FAIL 0; PHYSICAL_100K_GATE PASS; FINAL_LATCH PASS.
STM32F446RE 100K	36 / 99,934 / 30	ADC failures 0; state failures 0; overall ARM 100K PASS.
CH32V307 V2 8 Oct	174 / 99,667 / 159	STATE_FAILURES 0; INVALID_ACTIONS 0; OVERALL_RISCV_V2_100K PASS.
CH32V307 Diligence R1 9 Oct	36 / 99,934 / 30	Separate physical DAC/ADC campaign, not V2; 100K gate PASS.

All four action histograms sum to 100,000. They are not drawn from the frozen 30-case reference corpus; different plants, starting conditions and firmware mean that unlike histograms do **not** establish cross-architecture disagreement. [E9–E12]

Target	Physical path / declared protocol
ESP32-S3 final	B25K physical reference → GPIO3 ADC → native FARO → GPIO14 PWM → external 1 kΩ/100 µF RC → GPIO2 ADC feedback.
ARM STM32F446RE	DAC PA4/A2 → electrical connection → ADC PA0/A0; 100K completed in 24.937 seconds.
RISC-V V2 8 Oct	Native frozen structural gate and deterministic 100K qualification. No validated DAC-to-ADC electrical loop was established for this particular V2 checkpoint.
RISC-V Diligence R1 9 Oct	Separate physical PA4 DAC1 → PA5 ADC_IN5 path; TIM2 and scope measurements described below.

ARM 100K terminal values: reference **2,662**, ADC **2,343**, DAC **2,336**, leaving a measured **319-count** reference-to-ADC residual. The 9 October RISC-V physical loop also retained a nonzero residual. Structural parity is not equivalent to perfect plant tracking.

13 / Historical latency and accuracy: different boundaries

The figures below are physical measurements associated with different campaigns. Their methods, firmware, peripheral I/O and timing boundaries are not identical; they must not be presented as a controlled processor-speed ranking or a matched 30-case latency study. [E9, E10, E12]

Campaign / method	Min (µs)	Mean (µs)	Max (µs)	Boundary
ESP32-S3 final GPIO3 100K	24	29.525	30	Internal final route under ADC/OSH/PWM/RC setup; reported CPU 160 MHz.

Campaign / method	Min (µs)	Mean (µs)	Max (µs)	Boundary
ARM 100K 3 Oct	209.500	241.250	241.875	Cortex-M DWT counter; 16 MHz declared conversion; 3352 / 3860 / 3870 cycles.
RISC-V Diligence R1 9 Oct	281	356.843	361	TIM2 1 µs resolution; ADC-read to DAC-write wrapper, excluding UART and settling.

Independent oscilloscope observations

Platform	External interval	Interpretation
ESP32-S3	23.0 µs route; 660 ns action/selection	Two different dedicated builds/markers, not a distribution.
STM32F446RE	~226 µs instrumented pulse	Not identical to the DWT counter boundaries.
CH32V307 9 Oct	284 µs PA8 interval	One instrument cursor interval around the wrapper, not a per-case sample series.

P50, P95 or P99 values are not available for a common 30-case protocol, nor can they be derived from minimum, mean and maximum. Strict and conditioned tolerances are acceptance conditions rather than observed error distributions. No common three-target physical-accuracy ranking is claimed.

Separate R2 timing, retained for chronology

The unsuccessful R2 100K run recorded 25 / 29.329 / 39 µs (min/mean/max). Those observations belong to R2 and cannot replace the final GPIO3 physical campaign timing, 24 / 29.525 / 30 µs. The two build identities and run outcomes remain separate. [E2, E3]

14 / External instrumentation, board-level power and memory

Platform	Instrument / observation	Boundary and limitation
ESP32-S3	UNI-T UTD2052CL+ARG: representative 23 µs route and 660 ns action interval.	Different scope markers; not all-case distribution.
ARM	Same scope family: ~226 µs route pulse and observed output transition.	Pulse is an instrumented window; zero cursor difference in a captured frame is not zero physical latency.
RISC-V Diligence R1	Original oscilloscope BMP with 284 µs PA8 interval.	ADC-read to DAC-write wrapper; a single capture.
ARM input power	4.98 V, approximately 27–38 mA → 134–189 mW.	Complete NUCLEO board input; not isolated core power.
RISC-V input power	4.82 V, 10–11 mA → 48–53 mW.	Board readings/snapshots; not bandwidth-logged active-run energy.
ESP32 Step20 memory	368,691 B free internal heap; 16,774,624 B free PSRAM at recorded checkpoint.	Specific Step20 firmware/runtime, not universal FARO memory footprint.

No measured energy per decision is derived. Startup, peripherals, regulators and entire boards affect supply readings. The reports distinguish board power from isolated MCU/core power and preserve the original external capture contexts. [E9, E10, E12]

15 / Provenance reconciliation: what is closed

Link or claim	Current status	Grounded basis / limitation
Frozen 30-case source fixture across three lineages	VERIFIED SOURCE IDENTITY	ESP/ARM/RISC source fixture SHA; ARM recomputed 9 Nov.
Same 30 cases in four archival ELFs	CLOSED IN OWNER-RUN STATIC AUDIT	Exact whole-ELF hashes; symbol/section extraction; four 49,920-byte identical tables.
ESP32 final 12-item evidence-set digest	NEWLY RECOMPUTED: MATCH	Recovered original JSON + generator formula; independently reproduced SHA of ordered identifier/length records.
ESP32 R2 RAM-loaded image → its 30-line UART	DIRECT CONSOLE SEQUENCE	esptool load_ram plus physically captured R2 case output; BIN hash recorded in transcript.
ESP32 final runtime → exact final BIN loaded	STILL UNATTESTED AT BYTE LEVEL	Final manifest and runtime reported; raw final load_ram transfer with matching final BIN hash not recovered.
ARM 100K programming → later 100K PASS	HISTORICALLY DOCUMENTED	3 Oct combined programmer/console record naming 100K ELF, verify/reset then UART; 6 Oct hashes later.
RISC-V V2 HEX → exact executed memory bytes	STILL UNATTESTED AT BYTE LEVEL	HEX identity and original physical UART recorded; original same-session WCHISP verify/readback not recovered.
RISC-V 00:16 UART digest → 11:08 UART file	DISTINCT SESSIONS	00:16 digest cannot be reused for the 11:08 capture or another file with same name.
Same-boundary latency ranking of all three	NOT ESTABLISHED	Existing timings measured different operations, image versions and peripheral inclusions.

The distinct ARM `osh_arm_scope.elf` programmer record is an instrumentation session and cannot be used as proof that the archived parity/100K image was flashed. The RISC-V HEX file name `FLASH_100K.hex` was reused; only full SHA-256 disambiguates versions.

16 / What additional attestation could still establish

The outstanding tasks are precise and do not negate the physical trials:

• Archive the exact 9 November ELF extraction output, checker version and four full-file identities together for authorized reruns of the static table comparison.

• Locate a raw, final-GPIO3 ESP32-S3 RAM-loader console that identifies the final BIN hash and shows the transferred bytes and subsequent runtime; a different R2 loader cannot substitute for it.

• Preserve the R2 RAM-loader plus failed 100K outcome alongside the original 30 structural PASS lines; if an R2 ELF is subsequently identified, compare its compiled fixture independently of the final ELF.

• Keep ARM 3 October programming and runtime originals together, but label the later 6 October file hashes as archival identities rather than contemporaneous MCU readback certificates.

• Preserve the original RISC-V 00:16 UART (hash 6e01…b1d) and the 11:08 recovery UART as different recordings; an authorized same-session programmer log or device-memory readback would strengthen image attestation.

• Treat a matched case-by-case latency and physical-accuracy study as a separate future experiment, requiring common start conditions, identical measurement boundary and raw timing samples.

Any new board readback would establish the bytes present in the device **at the time of that new readback**. If another image has since been flashed, it cannot retroactively prove what ran in October. A new operation must be authorized separately; this report does not recommend reflashing original validated firmware merely to manufacture a historic link.

Most defensible public conclusion

> **Physical decision-reference portability has been demonstrated within the tested native FARO path.** The three MCU families report passing structural parity on a common 30-case reference, and a later static ELF audit reports identical compiled reference data across four identified archival images. The historical endurance and latency measurements remain distinct campaigns with clearly stated boundaries. The remaining image-to-session links are disclosed rather than assumed.

This result does not establish general intelligence, independent reproduction, safety certification, a fully closed 90-line direct UART triplet, bit-identical whole-program code, a common latency ranking or portability of every FARO subsystem.

A / Original ESP32-S3 R2 UART: all 30 individual results

These records are reproduced from the preserved 21 September R2 physical serial transcript. They are R2 output, **not** ARM/RISC-V UART lines and **not** substituted for the final GPIO3 run. The 30-case structural result passed; the R2 100K physical gate later failed. [E2]

CASES 01–15

```
CASE=01 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=3 SEL=3 FINAL=2 MEANING=c9bacc8972629cd6
CASE=02 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=03 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=04 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=05 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=06 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=07 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=08 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=09 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=10 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=11 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=12 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=13 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=14 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=15 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
```

CASES 16–30

```
CASE=16 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=17 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=18 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=19 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=20 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=21 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=22 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=23 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=24 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=25 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=26 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=27 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=28 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=29 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=30 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
```

U = original printed five-level PASS. NUM is the numeric-policy path; HAL/SEL/FINAL are selector counts, and MEANING is an anonymized structural digest. These are not analog INC/HOLD/DEC actions. Every original CASE line is preserved without changing values.

B / Physical UART acceptance excerpts and test identities

ARM parity / 20K - verified original capture

```
OSH_ARM_EXECUTION_BEGIN
BOARD=NUCLEO-F446RE
CPU=CORTEX-M4
PARITY_CASES=30
PARITY_EXECUTION_BEGIN
PARITY_EXECUTION_END
L1_L2_L3_L4_L5_ALL_30_CASES=PASS
PC_ESP32_PARITY_GATE=PASS
OVERALL_PAPER_CLOSEOUT=PASS
OSH_ARM_EXECUTION_END
```

RISC-V V2 later recovery - original timestamped record (excerpt)

```
2026-10-08 11:08:49.139-03:00 | BOARD=CH32V307VCT6
2026-10-08 11:08:49.147-03:00 | CPU_HZ=96000000
2026-10-08 11:08:49.155-03:00 | PARITY_CASES=30
2026-10-08 11:08:49.191-03:00 | STRUCTURAL_PARITY_30_30=PASS
2026-10-08 11:08:49.195-03:00 | GATE_5_OSH_STRUCTURAL_PARITY=PASS
2026-10-08 11:08:49.204-03:00 | ENDURANCE_CYCLES=100000
2026-10-08 11:09:23.412-03:00 | RISCV_100K_PROGRESS=100000/100000
2026-10-08 11:09:23.418-03:00 | STATE_FAILURES=0
2026-10-08 11:09:23.422-03:00 | INVALID_ACTIONS=0
2026-10-08 11:09:23.425-03:00 | ACTION_INC=174
2026-10-08 11:09:23.428-03:00 | ACTION_HOLD=99667
2026-10-08 11:09:23.430-03:00 | ACTION_DEC=159
2026-10-08 11:09:23.473-03:00 | OVERALL_RISCV_V2_100K=PASS
```

The 00:16 RISC-V original session is preserved in its separate historical record with an archival digest. The 11:08 excerpt here is from a distinct later recovery and is not assigned the earlier capture hash.

Owner-run 9 November table comparison: exact outcome labels

```
ARM_PARIDAD  30 cases  49920 bytes  SHA256=4a771957...7405157baf
ARM_100K    30 cases  49920 bytes  SHA256=4a771957...7405157baf
RISCV_V2    30 cases  49920 bytes  SHA256=4a771957...7405157baf
ESP32_FINAL 30 cases  49920 bytes  SHA256=4a771957...7405157baf
COTEJO_TABLAS=IDENTIDAD_BYTE_A_BYTE_30_CASOS
```

The displayed table-digest abbreviation is solely for compactness in this excerpt. The complete 64-character value appears in section 03. The checker’s full original terminal output should be preserved for independent static audit replay.

C / Complete non-confidential SHA-256 evidence register

Compiled 30-case table	4a77195717d3f87585af84f5e6c0d2d8 312ed3cbcb1aaf2f7984ab7405157baf
Frozen common fixture source	23863039f2b2b920fe578c6dc6e0e006 24ba34e6add00709522be8d6e70ec321
ESP final 12-item evidence-set digest	d38648587bc627409e770f5b3fd3b297 2e9abe31094db8550257fc8c9a6a0106
ESP R2 recorded BIN	4440dc05a6c4d39dac8b6223feb25303 c303fa28b774b6e03d7336ae9aed3811
ESP Step20 stability FINAL ELF	41bbe7034913664437e4204819df1fac b645767cb752807a207a207928c58540
ESP Step20 stability FINAL BIN	4b79d07b06fef50615b3ed54126d49b1 b2291be62445125bf5776cca23b7a0d8
ESP final GPIO3 BIN	86457e990bd1b96a093f418c5ce8f71b 60d41f55c4065757df4ca543c7b4863d
ESP final GPIO3 ELF	30cbbf62952b442b41aa2bf1398bd937 7bd1401a18a5da41a1ea74dd601f1ff3
ARM parity/20K ELF	9cd4c56ab5c29bb150900c96abff3328 fdeb1b28e13b639fd97fbb0f6a7a0645
ARM parity/20K BIN	336788116049fcfcbee03210c48b32ca b92c5670216e59765280c016440fcaae
ARM parity/20K UART	c58ec37b2995e00e1f71f66659d5c112 2d613fe1d959780c9524d5e5a4b63720
ARM 100K ELF	98a420f518d51bbd2d7a797d87df8b2c f825189181d1d7c53ada6ec1d2b95aa1
ARM 100K BIN	7ae036ef3983fdbf72081904de591632 fd830fc6834ee7aae47ab7d2eb212997
ARM 100K UART	7d40efbca2b4393c697f128bfce78b32 881692188b11aa0b798685e57e57ef11
RISC-V V2 100K ELF	cb9f88f38e755cb82f51fdfe0bf8000f f0963eb774195e1b3b965ae1975c1517
RISC-V V2 100K HEX	544338ec6e5a4fa4251d8dea7f557074 418c8014ea297fd2cebaa02f9134ad21
RISC-V V2 00:16 original UART	6e01a89c2d17af83c2527e549e142038 ae0e66816721b8eaea2b45cc454b7b1d
RISC-V Diligence R1 9 Oct HEX	2db7f4dba1893c67be1ebcdca9f0b0a6 db43484eb9f1fb75e45301bf63bb1268
Static ELF checker source as retained	e96f196091e81f9aa5ba2abafc35d352 f8637693a21e12fdc5e42cd92e398822

Some digests were recomputed locally by the owner during previous sessions; others are values transcribed from primary manifests or dated reports. The common manifest digest was independently reproduced from the recovered recorded list and its generating algorithm in this revision. Hashes of private original firmware bytes were not all recomputed by this report author.

D / Original 12-artifact final ESP32-S3 evidence manifest

This table reproduces every label, byte length and original SHA-256 in the recovered final GPIO3 manifest. Workstation paths are intentionally excluded. The register order is preserved because the evidence-set digest is order-sensitive. [E1]

Manifest label	Bytes	SHA-256 recorded in original manifest
final_main	29910	331c7c5298acfcc5a9d8020af6a036e7 d77f253ddcebf0276f9d6afbd67c5000
core	5271	e09f1d7928f291a64ab5c8a8d6f9af62 cb2b293a1022e6897044fd04288fd363
hal	14810	da9fa7a321a6bf1df797054f9dcac132 349930eef8aa2ba11e3c078a188f422f
vortex	2922	0c405d2e86ec9dfbb54b03363d984bb8 c745ab84cfdafdab94205351a7e73d78
selector	23967	26c4971a12209460ab74e188c4c234f5 bd80db17ac15167ca3802faecd2b630b
compact_graph	15828	fe5dcaa69ec4cc1d9a111a4ad8cd6005 7ddd4a9bbf63984167d4c7b7beef80eb
project_cmake	203	42e7309253a2c1ae697c59f3906824fb 1ce5baac5b285017bafbe62624b634d4
main_cmake	334	3db2213694f18e9609ef1e14b2e0a6cd 6da45f8cac7e5e1e55e08fc59df5b047
sdkconfig_defaults	600	f9a74d47c1bab31699995b87c24c9c5d 3ace5eff39d8a35a97d773273aca8a29
sdkconfig	68226	67b563785c07513a1fe47234fdc874bb 939330cabbfdb7962f4a3fcf768ddbd6
firmware_bin	202784	86457e990bd1b96a093f418c5ce8f71b 60d41f55c4065757df4ca543c7b4863d
firmware_elf	3432132	30cbbf62952b442b41aa2bf1398bd937 7bd1401a18a5da41a1ea74dd601f1ff3

Reproduction of the set fingerprint

The original manifest generator hashes the UTF-8 bytes of the ordered concatenation `label|sha256|byte_length`, joined by the literal two characters backslash and n between records. This edition independently reproduced the reported digest using only the twelve manifest entries and the original generator formula:

Recomputed evidence-set SHA-256	d38648587bc627409e770f5b3fd3b297 2e9abe31094db8550257fc8c9a6a0106

Verified conclusion: exact consistency between the recovered twelve entries, the generator formula and its recorded set digest. No assertion is made that original source/BIN/ELF bytes were all separately rehashed for this edition, or that a particular image was loaded at a historic timestamp.

E / Chronology, source register and review boundaries

ID	Primary or secondary evidence and provenance
E1	Original 21 September ESP32-S3 final 12-artifact SHA-256 TXT/JSON evidence manifests and their original generating script; exact manifest digest independently recomputed in this revision.
E2	Original September ESP32-S3 R2_MEANING_FIX host/MCU transcript; esptool load_ram, 30 printed cases, R2 100K FAIL, later physical latch FAIL. Direct records reviewed.
E3	Recovered final ESP32-S3 GPIO3 runtime described in the owner-supplied 9 November provenance addendum; final 30-case gate, final 100K PASS, reported build/clock. Original complete final load_ram transcript not independently recovered.
E4	Original ARM parity/20K UART archive, exact SHA-256 recomputed on 9 November and compared to the dated original ARM validation report.
E5	3 October ARM programmer and 100K UART sequence described in recovered primary transcript and the ARM physical migration roadmap; separate 6 October archival hash output.
E6	8 October 00:16 RISC-V V2 ELF/HEX hashes, original timestamped UART and immediate hash output, described in recovered historical evidence and checkpoint.
E7	Separate 8 October 11:08–11:09 RISC-V V2 owner-supplied timestamped UART recovery capture, with full runtime gates and actions.
E8	9 November owner-run four-ELF read-only checker: 54 candidates, exact ELF hashes, OSH_PARITY_CASES extracted, identical 49,920-byte tables; checker source inspected in this edition.
E9	OSH Physical Validation of an Autonomous Homeostatic Cognitive Architecture, 21 September 2026 (17-page physical ESP32-S3 research paper).
E10	OSH ARM Physical Validation, 3 October 2026 (physical ARM qualification, UART, scope and hash register).
E11	OSH RISC-V Technical Physical Validation Checkpoint, 8 October 2026 (V2 acceptance, timestamps, separate V3 non-validation).
E12	OSH RISC-V Public Benchmark Complete, 9 October 2026 (separate native DAC-to-ADC loop, timings, board power and host-controller comparisons).
E13	FARO 30-Case Source Parity Reconstruction, 8 November 2026 (source-level predicates, fixture hashes and 11 anonymized classes).
E14	Owner-supplied FARO Cross-Architecture Evidence and Recovered Provenance addendum, 9 November 2026. Recovered originals described; original bytes not all directly re-inspected in this revision.
E15	FARO Cross-Architecture Complete Parity, 9 November 2026. Previous 15-page consolidated report, data and UART appendix retained and corrected in this integrated edition.

Selected public publication URLs provided by the owner

www.oshsystems.com — main evidence portal

oshsystems.com/FARO_CROSS_ARCHITECTURE_COMPLETE_PARITY_EN.pdf — prior consolidated parity publication location as supplied; live status not independently checked.

oshsystems.com/osh_global_systems_vallejos_2026.pdf — ESP32-S3 original physical paper link as supplied.

oshsystems.com/assets/FARO_Memoria_ELF_Verificada_20261105.pdf — embedded memory ELF evidence link as supplied.

External reproducibility requires authorized access to original binaries, raw UART, manifest files and programmer logs. These materials are not automatically released by publishing this document. Source ownership and black-box confidentiality remain with OSH. Historical timestamps are not cryptographic notarization; missing exact-time events are not synthesized retrospectively.

VOLUME III

Recovered Provenance Addendum - Full Text

All ten owner-supplied sections, references and archival source notes. This update is retained intact and is not replaced by a summary.

Original source pages follow without alteration. A new section does not supersede or erase the retained record.

Recovered Historical Evidence and Provenance

Owner-supplied text of 9 November 2026. Preserved as a separate evidentiary layer, without silently replacing earlier records.

FARO Cross Architecture Evidence and Recovered Provenance
Addendum to the physical parity report of 9 November 2026
OSH Global Systems Inc

1. What this addendum establishes

Recovered historical records strengthen the correspondence between the reported physical results and the archived FARO artifacts. In particular, the final ESP32-S3 campaign contains its own 30-case structural parity output followed by the 100,000-cycle physical run. The final campaign therefore no longer has to rely solely on the earlier R2 parity record for its reported 30-case result.

The recovered originals strengthen artifact identities and historical sequences already described in the public report. They also distinguish three ESP32-S3 artifact groups, substantiate the full R2 BIN identifier and RAM-load transcript, and preserve the unsuccessful R2 100K outcome separately from the later successful campaign. These are substantive additions to the evidence chain. They do not establish a complete, independently reproducible chain from every archived executable to every historical execution.

This addendum concerns the execution and portability of a native FARO decision path. FARO is an embedded artificial intelligence architecture; the physical control experiments are applications of that architecture. These experiments alone do not validate all of FARO's cognitive capabilities, general intelligence, or suitability for safety-critical deployment. No new physical experiment is reported here.

2. Evidence classification and interpretation

The review distinguishes directly recovered primary records, later documentary references, and links still requiring verification. A primary console record documents what was recorded in that session. An original manifest documents the artifact identifiers recorded by its author. A later report or description of an image is secondary evidence unless the underlying artifact has also been examined.

The dates of recovered messages establish when material was submitted to the archive. They are not substitutes for instrument timestamps. File modification times and build strings provide context but are not authenticated proof of when a test occurred. Hashes quoted from a manifest or console record are identified as recorded values; quoting them does not imply that every corresponding binary was independently rehashed during this review.

3. Common fixture in the four archived executables

The audit reported on 9 November identifies an OSH_PARITY_CASES object of 49,920 bytes, corresponding to 30 records of 1,664 bytes, in four archived ELF executables. The reported extraction uses the symbol address and the associated ELF section rather than treating the symbol address as a file offset. The extracted object has the same reported SHA-256 in all four files:

```
4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf
```

The four full-file identifiers reported for the audited executables are:

ARM parity and 20K ELF
9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645

ARM 100K ELF
98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1

RISC-V 100K V2 ELF
cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517

ESP32-S3 final GPIO3 ELF
30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

This is evidence of a common embedded fixture in these archived executables. It is not a claim that architecture-specific machine code is byte-identical, nor that the fixture hash alone authenticates which executable ran on a device. The extraction script, raw audit output, and access to the underlying artifacts remain the basis for independent repetition of that static audit.

4. ESP32-S3 final physical campaign

The recovered final runtime record, submitted on 21 September 2026 at approximately 21:02 UTC, includes 30 individual case results covering the five structural levels, followed by a 100,000-cycle physical execution and a passing overall gate. The record identifies ESP-IDF v5.4.4 and a CPU frequency of 160 MHz. Its compile string is 21 September 2026 at 17:20:03, with no verified timezone attached to that build string.

The five levels cover endogenous final state, settle or vortex behavior, HAL candidates and selection, exact selector decisions, and semantic outputs. A reported pass is interpreted according to the verifier and tolerances of that campaign. It is not a claim that every floating-point intermediate is bit-identical across processors.

The final runtime declares:

```
EXECUTION=RAM_ONLY
MOTHER_FLASH_WRITE=FALSE
```

The relevant historical load operation is therefore a RAM-load operation, not programming of persistent application flash. A flash write or flash readback log must not be substituted for the missing RAM-loader evidence of this campaign. The archive includes an instruction to use esptool load_ram, but a command instruction is not proof that the command was executed.

Two original evidence manifests dated 21 September 2026, one TXT and one JSON, consistently identify 12 artifacts for OSH_ESP32S3_PAPER_CLOSED_LOOP_FINAL_GPIO3_V1. They record:

Final BIN: 202,784 bytes
SHA-256: 86457e990bd1b96a093f418c5ce8f71b60d41f55c4065757df4ca543c7b4863d

Final ELF: 3,432,132 bytes
SHA-256: 30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

Recorded EVIDENCE_SET_SHA256:
d38648587bc627409e770f5b3fd3b2972e9abe31094db8550257fc8c9a6a0106

The recovered manifests are primary documentary evidence for this artifact pair. Their values agree with the final pair cited in the public ESP32-S3 paper. The set digest has not been independently recomputed in this review, and its construction must be specified before another reviewer can reproduce it.

The remaining limitation is precise: the runtime's ELF hash line contains a zero placeholder rather than an authenticated full executable digest. A raw final-GPIO3 RAM-loader record binding the actual loaded BIN to the recorded digest has not yet been recovered. The manifests and runtime record strengthen campaign identification, but do not by themselves close that last cryptographic link.

5. ESP32-S3 artifact reconciliation

The recovered evidence supports distinguishing artifact groups instead of treating every differing hash as a conflict for one file.

Step20 stability final
The public memory report identifies OSH_ESP32_STEP20_STABILITY_V1_FINAL.elf by:
41bbe7034913664437e4204819df1facb645767cb752807a207a207928c58540
Its associated BIN is reported as 287,792 bytes with:
4b79d07b06fef50615b3ed54126d49b1b2291be62445125bf5776cca23b7a0d8
This is not the final GPIO3 artifact pair.

R2 meaning fix campaign
An original loader and runtime transcript submitted on 21 September 2026 at 02:00:23 UTC identifies the R2_MEANING_FIX BIN as 202,784 bytes with the full recorded SHA-256:
4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811

The same primary text records an esptool.py v4.12.0 load_ram operation and the outcome All segments done, executing at 40375c20. This primary source substantiates the full R2 identifier already printed in sections 04 and 10 of the base report and resolves the shortened 4440dc05... reference in a separate secondary addendum. It identifies an R2 artifact and an actual recorded RAM-load operation, not the Step20 BIN and not the final GPIO3 BIN. The digest is recovered from recorded hash output, rather than newly recomputed from the binary during this review.

The R2 output reports all 30 cases passing L1 to L5, but PHYSICAL_100K_GATE=FAIL. The 100K decision counts are increase 0, hold 100,000, decrease 0. Its reported latency is minimum 25, mean 29.329, and maximum 39 microseconds. Those outcomes must remain intact; neither its 100K result nor its latency may be substituted for the final GPIO3 campaign. The raw transcript also contains a preceding Step20 startup and watchdog events, requiring segmentation by load and boot boundary rather than treating the entire file as one execution.

Final GPIO3 physical campaign
The original manifests identify the final ELF and BIN as the 30cbbf62... and 86457e99... pair set out above. The recovered final runtime contains the 30-case result and 100K run.

The distinction between these groups resolves an important naming ambiguity. It does not establish a complete build history, an independently checked source diff, or that one group formally superseded every other group. Reused labels such as R2 must be qualified by project, artifact identity, and campaign.

6. ARM 100K programming and execution

A recovered primary console transcript submitted on 3 October 2026 at approximately 19:21 UTC records STM32CubeProgrammer v2.23.0, the NUCLEO-F446RE target, programming of osh_stm32_100k.elf, successful verification, software reset, and the subsequent UART parity and 100K results. It contains the sequence reported as Download verified successfully, MCU Reset, and Software reset performed. The runtime reports a passing 30-case gate, 100,000 cycles, and a passing overall result.

This directly strengthens the historical programming-to-runtime narrative. It is more specific than an unnamed statement that an ARM image was programmed successfully. A second programming record on the same date is a separate session and is not merged into the first sequence.

Primary hash output recovered from 6 October identifies the ARM 100K ELF by:
98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1

The later frozen set also records:
BIN: 7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997
UART: 7d40efbca2b4393c697f128bfce78b32881692188b11aa0b798685e57e57ef11

The 6 October hash evidence establishes the later archived identities. It is not a contemporaneous digest printed inside the 3 October programming operation. The recovered console transcript also does not supply authenticated timestamps for every write, verify, reset, and runtime event. The original raw programmer log and a device readback are not claimed to have been recovered by this review.

osh_arm_scope.elf is a distinct instrumentation image. It must remain separate from osh_stm32_100k.elf and from the ARM parity or 20K image. Scope observations cannot be used to authenticate the 100K executable.

7. RISC-V V2 programming and execution

Primary hash output in the recovered V2 sequence identifies:

ELF: cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517
HEX: 544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21

A primary timestamped UART record covers 8 October 2026 from 00:16:16.097 to 00:16:50.425 at UTC-03. It identifies CH32V307VCT6, QingKe V4F, and 96 MHz. It reports 30/30 parity, 100,000 cycles, zero state or invalid-result failures, a passing overall gate, and CAPTURE_COMPLETE=YES. Immediately subsequent hash output identifies that captured UART artifact as:

```
6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d
```

The archive also contains a description of a programming screenshot reporting successful erase, programming, and verification shortly before that capture. The underlying screenshot was not independently re-inspected in this recovery, and its description is not a raw programmer log or a device readback. A preceding precheck declares WCHISPTool V4.00 and USB bootloader 02.90; this does not independently attest the tool version used in every subsequent programming session.

A later recovery run around 11:08 to 11:09 repeats the aggregate outcome. It is a separate execution. The UART filename was reused, so the hash for the 00:16 capture must not be assigned to a later file solely because its filename is unchanged. FLASH_100K.hex was also reused for a later image with a different digest, while the frozen V2 HEX retained the 544338ec... identity.

The recovered evidence therefore supports a specific V2 runtime record and its recorded file identity. A hash-bound V2 device readback remains unestablished. Later programming evidence for the Diligence R1 image does not close the V2 chain. If an additional device session is eventually authorized, it can attest that new session; it cannot retroactively generate evidence of what was loaded in an earlier one.

8. Results that remain valid and limits on comparison

The recovered campaigns report these 100K decision counts:

ESP32-S3 final: increase 5,160; hold 93,358; decrease 1,482.
ARM 100K: increase 36; hold 99,934; decrease 30.
RISC-V V2: increase 174; hold 99,667; decrease 159.

Each sums to 100,000. These campaigns did not establish an identical physical plant input sequence, so differences in these totals are not a direct measure of cross-platform decision disagreement.

Historical latency values must also retain their campaign boundaries:

ESP32-S3 final, 160 MHz: minimum 24 microseconds, mean 29.525 microseconds, maximum 30 microseconds for the reported internal route over 100K samples.
ARM 100K, 16 MHz: minimum 209.500 microseconds, mean 241.250 microseconds, maximum 241.875 microseconds, corresponding to reported DWT cycle values 3,352, 3,860, and 3,870.
RISC-V Diligence R1, 96 MHz: minimum 281 microseconds, mean 356.843 microseconds, maximum 361 microseconds for the reported ADC-read-to-DAC-write interval, including the wrapper and excluding UART and external settling.

The RISC-V latency campaign is Diligence R1, not the V2 100K campaign described above. The reported intervals and instrumentation are not a single matched benchmark. They cannot support a speed ranking, cross-platform energy estimate, or claims of equivalent latency measurement boundaries. Medians and percentiles cannot be reconstructed from minimum, mean, and maximum alone. Scope measurements remain separately identified observations rather than replacements for latency distributions.

9. What is established and what still needs closure

The combined documentary record supports native physical execution on the three platforms, the reported structural parity gates, campaign-specific endurance results, and the reported identity of the common fixture in the four archived executables. The recovered ESP final record specifically strengthens the association of the final physical campaign with a complete 30-case structural gate.

The remaining closure items are bounded:

1. Preserve the raw 9 November extraction output and its exact script version together with the four artifact identifiers, so the static fixture audit can be repeated under authorized access.
2. Recover the exact final ESP32-S3 RAM-loader command and raw output, if retained, and bind them to the final BIN identifier. Do not replace this with a flash-programming record or with a loader record from another campaign.
3. Preserve the recovered R2 loader and runtime transcript with its full recorded BIN digest. The full R2 identity already cited in the base report is now supported by the recovered primary transcript. Extraction of the R2 BIN fixture and a comparison with the final ELF fixture remain separate, unestablished checks. Retain the R2 100K failure alongside its passing 30-case structural result.
4. Preserve the ARM programming and UART originals, explicitly distinguishing the 3 October execution

from the later 6 October archival hashes. Do not fabricate missing event timestamps.
5. Preserve the RISC-V 00:16 UART by hash and keep later recovery files separate. A device readback for the V2 image has not been established. Any new programming or readback session requires a separately authorized procedure.
6. Treat a common-boundary latency study as a separate experimental question. It is not a prerequisite for accurately reporting the existing fixture and parity evidence, but it is required for a matched latency comparison.

10. Evidence preservation and disclosure

The evidence index should identify each original artifact, its byte length, its full SHA-256, its campaign, and the operation that produced it. A later erratum should remain separate from the preserved original. A redacted disclosure copy is a derivative artifact and must have its own identifier and hash.

A hash establishes byte identity relative to the stated reference. It does not by itself prove execution, acquisition time, authorship, or an independent chain of custody. Publication dates and trusted timestamping can strengthen chronology when available, but must not be reconstructed retrospectively.

Public review does not require automatic disclosure of private firmware, source code, memories, or complete internal traces. Those artifacts remain subject to the owner's explicit authorization and an appropriate review arrangement. This addendum records the evidence state without authorizing publication or transfer of protected materials.

References

[1] FARO Cross Architecture Complete Parity report, 9 November 2026. Public report and self-audit. https://oshsystems.com/FARO_CROSS_ARCHITECTURE_COMPLETE_PARITY_EN.pdf

[2] OSH Global Systems, Vallejos, 2026. ESP32-S3 physical execution report. See the final artifact pair recorded in the evidence section.
https://oshsystems.com/osh_global_systems_vallejos_2026.pdf

[3] FARO Memoria ELF Verificada, 5 November 2026. Step20 artifact identification, page 3.
https://oshsystems.com/assets/FARO_Memoria_ELF_Verificada_20261105.pdf

[H1] Original final ESP32-S3 TXT and JSON evidence manifests dated 21 September 2026, together with the recovered final runtime record from that date.
[H2] ARM 100K programming and runtime console record submitted on 3 October 2026, and primary archival hash outputs submitted on 6 October 2026.
[H3] RISC-V V2 primary ELF and HEX hash output, timestamped UART capture of 8 October 2026 at 00:16 UTC-03, and its immediately subsequent SHA-256 output. Later recovery sessions are indexed separately.
[H4] Original R2_MEANING_FIX RAM-loader and runtime transcript submitted on 21 September 2026 at 02:00:23 UTC. This primary record supports the full R2 identity in the base report and resolves the shortened reference in a separate consolidated evidence addendum dated 9 November 2026, page 2.

Historical records H1 to H4 are archival sources, not represented here as publicly downloadable artifacts. Independent reproduction requires authorized access to the relevant originals.

VOLUME IV

Reproduced Digest and Verification Register

New supporting verification annex. Recalculates the final ESP32-S3 12-item evidence-set digest and prints every hash continuously.

Original source pages follow without alteration. A new section does not supersede or erase the retained record.

Independently Reproduced Manifest Digest and Integrity Register

Review performed on the recovered original 12-artifact TXT/JSON ledger and its saved generator algorithm. This appendix is new verification evidence; no original BIN/ELF data or device memory was obtained or altered.

1 / Reproducible SHA-256 evidence-set formula

Recovered generator order: 12 source manifest entries, each expressed as label|sha256|byte_length. Entries are joined with the two literal characters backslash and n (\n), rather than an actual newline, encoded as UTF-8 and hashed with SHA-256. The order of entries matters.

Recorded evidence-set SHA-256

`d38648587bc627409e770f5b3fd3b2972e9abe31094db8550257fc8c9a6a0106`

Independently recalculated evidence-set SHA-256

`d38648587bc627409e770f5b3fd3b2972e9abe31094db8550257fc8c9a6a0106`

RESULT: EXACT MATCH. This demonstrates internal consistency of the 12-item manifest and reproducibility of its listed-data digest. It is not a rehash of all underlying firmware/source files, nor proof of which BIN ran at the historical time.

2 / All 12 original final ESP32-S3 manifest entries

Each recorded length and digest below is taken from the recovered original ledger. Hashes are complete and continuously printable; workstation paths are intentionally omitted.

01 final_main · 29,910 bytes

`331c7c5298acfcc5a9d8020af6a036e7d77f253ddcebf0276f9d6afbd67c5000`

02 core · 5,271 bytes

`e09f1d7928f291a64ab5c8a8d6f9af62cb2b293a1022e6897044fd04288fd363`

03 hal · 14,810 bytes

`da9fa7a321a6bf1df797054f9dcac132349930eef8aa2ba11e3c078a188f422f`

04 vortex · 2,922 bytes

`0c405d2e86ec9dfbb54b03363d984bb8c745ab84cfdafdab94205351a7e73d78`

05 selector · 23,967 bytes

`26c4971a12209460ab74e188c4c234f5bd80db17ac15167ca3802faecd2b630b`

06 compact_graph · 15,828 bytes

`fe5dcaa69ec4cc1d9a111a4ad8cd60057ddd4a9bbf63984167d4c7b7beef80eb`

07 project_cmake · 203 bytes

`42e7309253a2c1ae697c59f3906824fb1ce5baac5b285017bafbe62624b634d4`

08 main_cmake · 334 bytes

`3db2213694f18e9609ef1e14b2e0a6cd6da45f8cac7e5e1e55e08fc59df5b047`

09 sdkconfig_defaults · 600 bytes

f9a74d47c1bab31699995b87c24c9c5d3ace5eff39d8a35a97d773273aca8a29

10 sdkconfig · 68,226 bytes

67b563785c07513a1fe47234fdc874bb939330cabbfdb7962f4a3fcf768ddbd6

11 firmware_bin · 202,784 bytes

86457e990bd1b96a093f418c5ce8f71b60d41f55c4065757df4ca543c7b4863d

12 firmware_elf · 3,432,132 bytes

30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

3 / Whole-ELF and compiled-case identity anchors

Common source fixture header

23863039f2b2b920fe578c6dc6e0e00624ba34e6add00709522be8d6e70ec321

Compiled 30-case ELF table (49,920 bytes)

4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf

ARM parity/20K ELF

9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645

ARM 100K ELF

98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1

RISC-V V2 100K ELF

cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517

ESP32-S3 final physical ELF

30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

ESP32-S3 R2 reported BIN

4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811

ESP32 Step20 stability ELF

41bbe7034913664437e4204819df1facb645767cb752807a207a207928c58540

ESP32 Step20 stability BIN

4b79d07b06fef50615b3ed54126d49b1b2291be62445125bf5776cca23b7a0d8

ARM parity/20K UART

c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720

ARM parity/20K BIN

336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae

ARM 100K BIN

7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997

ARM 100K UART

7d40efbca2b4393c697f128bfce78b32881692188b11aa0b798685e57e57ef11

RISC-V V2 100K HEX

544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21

RISC-V V2 00:16 UART

6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d

RISC-V Diligence R1 (9 Oct) HEX

2db7f4dba1893c67be1ebcdca9f0b0a6db43484eb9f1fb75e45301bf63bb1268

Original compiled-table audit checker SHA-256 (archived)

`e96f196091e81f9aa5ba2abafc35d352f8637693a21e12fdc5e42cd92e398822`

4 / Chronology and closure classification

• ESP32-S3 R2: original esptool load_ram transcript records its own BIN digest and direct 30-case UART PASS; the R2 100K gate failed and is preserved. This R2 image is separate from final GPIO3.

• ESP32-S3 final GPIO3: original final-runtime 30-case PASS and final 100K PASS, plus a 12-item ELF/BIN manifest. The final RAM-loader transcript hash-bound to that exact BIN has not been recovered.

• ARM parity/20K: original UART digest locally reverified, PASS at 30 cases. The archived ELF table has been compared with the four-image corpus.

• ARM 100K: 3 October programmer/console sequence names osh_stm32_100k.elf and reports verify/reset followed by parity and 100K PASS. Later 6 October full-file hash is not a contemporaneous device readback.

• RISC-V V2 00:16: original 8 October time-stamped UART and immediate SHA-256 are reported, separate from a later 11:08 recovery capture; V2 HEX identity is archived. A hash-bound readback proving the exact loaded bytes was not recovered.

• RISC-V Diligence R1 9 October: distinct firmware and physical DAC-to-ADC experiment, with distinct decision counts and latency boundary; do not substitute its timing for RISC-V V2.

• Three platform latency means are historical measurements with different boundaries; they are not a synchronized three-target latency ranking.

• No private executable or source is released by this public archive, and this document records no new hardware execution.

5 / Primary evidence versus documentary reconstruction

This preservation master includes three levels of material: (a) historical device output and first-party SHA-256 manifests, (b) owner-run static ELF extraction and later local hash checks, and (c) documentary interpretation. A checksum attests to recorded-byte identity relative to the reference; it does not by itself attest historical program load, device state, or independent replication. The original 15-page and integrated 18-page documents are preserved unchanged; their overlap is archival preservation, not evidence of separate experiments. In case of discrepancy, review each original session and the addendum evidence classification rather than retroactively editing the records.